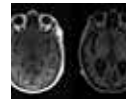


**Carbon tax in US?**

The White House and at least 49 senators support a proposal to impose carbon tax.
<https://digit.in/dec21-13>

**Cure Alzheimer's with a jab**

Alzheimer's cure breakthrough as jab could restore patients' memories.
<https://digit.in/dec21-14>

Earth Management with Landsat 9

Space data to the rescue!

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y 2020, 114 flights launched approximately 1,300 satellites into orbit, breaking the 1,000

new satellites per year barrier for the first time. However, in 2021, around 1,400 new satellites have already begun orbiting the Earth as of September 16, with the number expected to rise as the year progresses. Due to exponential evolution in the electronics and mechanics supporting a satellite, it has become easier to launch one. The vast majority of aircraft launched in 2020 were smallsats, with a weight of 600 kilograms or less.

However, some aircraft which carry heavy instrumentation with them weigh a lot more. These satellites aim to collect vital information regarding the Earth's atmosphere and climate, and these instruments contribute to the overall launch mass of a satellite. Keeping all these satellite design frameworks in mind, Northrop Grumman Innovation



Systems designed the latest addition to the Landsat series: Landsat 9, the ninth in the Landsat program which will continue the role of monitoring and managing land resources.

WHAT IS THE LANDSAT PROGRAM?

Landsat is the world's longest-running continuous collection of moderate-resolution land remote sensing data from space. Overall, the program is a strategic element that helps the US Department of the Interior manage Federal lands effectively. Landsat data is used for research, industry, education, and other purposes all across the world.

NASA and the US Geological Survey jointly operate the Landsat

Program, which consists of a series of Earth-observing satellite missions. The Earth Resources Technology Satellite (ERTS-1) was launched on July 23, 1972, in collaboration with NASA. It was dubbed Landsat 1 subsequently. In the 1970s and 1980s, more Landsat satellites were launched. In 1999, Landsat 7 was launched, followed by Landsat 8 on February 11, 2013. After 8 long years, Landsat 9 has left the Earth to follow, and eventually replace, Landsat 7. Since Landsat 9 joins Landsat 8 in its data acquisition, it is largely built on the same framework as its predecessor. This reduces the build time and risk of gaps in observations.

SPRUCING UP THE LANDSAT ARMORY WITH LANDSAT 9

In October 2016, NASA granted a delivery order contract to Northrop Grumman for the design and construction of Landsat 9, as they were responsible for the complete construction and testing of the previous version as well. The US\$129.9 million project cost is based on a five-year agreement between the two compa-

